



Predicting Six-Year Graduation Rates at U.S. Four-Year Colleges

Capstone Final Presentation

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Executive Summary

- ▶ Objective: explain and predict pooled six-year graduation at operating, bachelor's-predominant U.S. colleges.
- ▶ Data: 1,669 eligible institutions from the June 10, 2026 College Scorecard release.
- ▶ Method: OLS with HC3-robust inference, five-fold cross-validation within the training data, and a 334-college holdout.
- ▶ Result: random forest achieved MAE 6.72 points, RMSE 9.13 points, and $R^2 = .768$.
- ▶ Importance: completion affects students, family investment, institutional planning, and public accountability.
- ▶ Use: a documented institutional benchmark for local review—not ranking, causation, or student scoring.

Gap Analysis

- ▶ Completion reflects financial need, price, selectivity, resources, staffing, control, region, and locale at the same time.
- ▶ Earlier studies often focus on students, one policy setting, or explanation without a national unseen-college test.
- ▶ Raw rankings mix institutional mission and student composition with performance.
- ▶ This study combines robust adjustment, four-model comparison, held-out prediction, uncertainty, subgroup checks, and reproducible outputs.

See Tinto (1975); Titus (2004, 2006); Matz et al. (2023).

Research Questions

RQ1: Affordability

After accounting for enrollment, admission rate, instructional expenditure per full-time-equivalent student, faculty salary, student-faculty ratio, control, region, and locale, what adjusted relationship do average net price, Pell recipient share, and federal-loan share have with pooled six-year graduation?

Hypotheses: H_0 : all three affordability coefficients equal zero. H_a : at least one coefficient differs from zero.

Result: Reject H_0 , $p < .001$.

RQ2: Control and Locale

Do the covariate-adjusted pooled six-year graduation means differ among institutional control groups or among locale groups?

Hypotheses: H_0 : adjusted means are equal within each factor. H_a : at least one group mean differs.

Result: Reject H_0 for control; do not reject H_0 for locale.

RQ3: Prediction

When applied to colleges excluded from training, how accurately do OLS, elastic net, random forest, and gradient boosting predict pooled six-year graduation?

Hypotheses: H_0 : predictions have no positive relationship with observed rates and do not reduce error relative to the training-mean baseline. H_a : at least one model has $r > 0$ and lower error.

Result: All four models beat the baseline; random forest was selected.

RQ4: Added Predictive Value

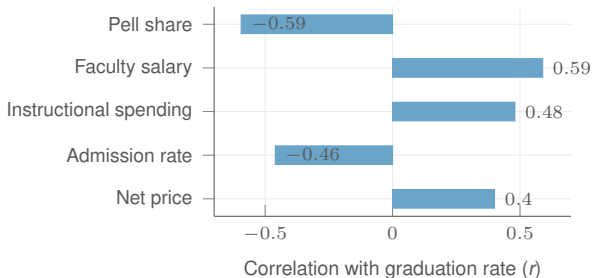
How much predictive value is gained when average net price, Pell share, and federal-loan share are added to the organizational and resource characteristics?

Hypotheses: H_0 : affordability produces no holdout R^2 gain and no error reduction. H_a : the complete feature set raises R^2 or lowers error.

Result: The full model reduces RMSE by 18.42%.

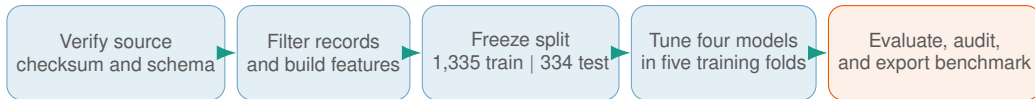
Data Description and EDA

- ▶ 6,273 raw Scorecard records → 1,669 eligible four-year colleges
- ▶ Unit: institution; outcome: pooled six-year completion
- ▶ Mean completion: 55.27%; the distribution is broad
- ▶ Train/test: 1,335 / 334
- ▶ Pell share and faculty salary tie for the largest absolute raw correlations shown: $|r| = .59$ (rounded)



Direction and length show sign and relative strength; adjusted results follow.

Architecture Diagram / Workflow



Key: blue = preparation/model selection | orange = holdout evaluation begins

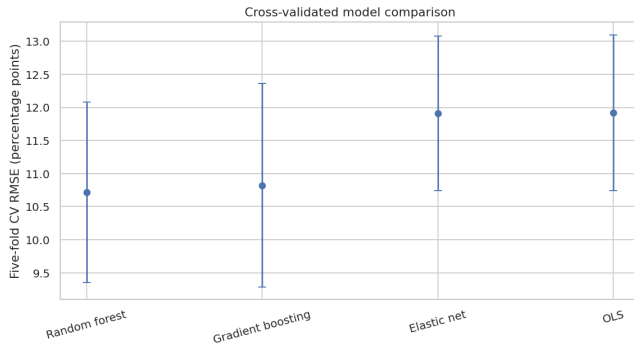
Fold-specific imputation and encoding prevent validation and test leakage.

Model Building: Candidate Models

- ▶ OLS: transparent linear benchmark with directly interpretable coefficients.
- ▶ Elastic net: regularized linear model for overlapping predictor information.
- ▶ Random forest: averaged trees for nonlinear relationships and interactions.
- ▶ Gradient boosting: sequential tree ensemble as a second nonlinear candidate.
- ▶ Selection rule: lowest mean five-fold training RMSE; one common 334-college holdout.

Methods: Breiman (2001); Friedman (2001); Zou and Hastie (2005).

Model Building: Cross-Validation and Selection



Error bars show ± 1 SD across five folds.

Selection was recorded before the 334-college holdout was opened.

Results: RQ1 – Affordability

- ▶ Joint HC3 test: $F(3, 1643) = 59.80$, $p < .001$; partial $R^2 = .160$.
- ▶ Pell share: $b = -.346$, $p < .001$; a .10 increase corresponds to a 3.46 percentage-point decrease.
- ▶ Log net price: $p = .074$; federal-loan share: $p = .497$.
- ▶ Pell share marks the concentration of financial need across colleges; it is not an individual Pell effect.
- ▶ Sensitivity checks preserve the negative Pell-share direction.

HC3 OLS results from the final analytic sample, $n = 1,669$.

Results: RQ2 – Control and Locale

Control

$F(2, 1643) = 37.80, p < .001$

Private nonprofit: 58.37%

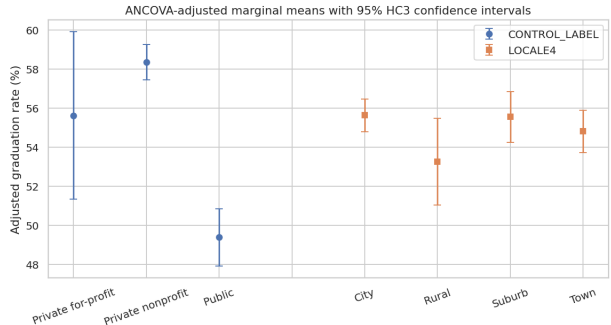
Public: 49.39%

Nonprofit – public: +8.98 points

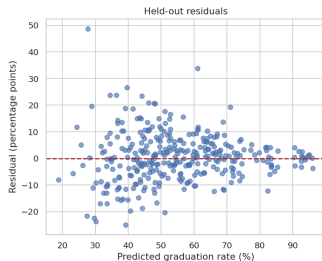
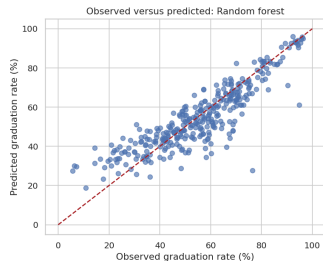
Locale

$F(3, 1643) = 1.46, p = .225$

All Holm-adjusted intervals include zero.



Results: RQ3 – Holdout Performance



MAE 6.72 points

RMSE 9.13 points

$$R^2 = .76769$$

51.81% lower RMSE than
mean baseline

Common frozen holdout: $n = 334$ institutions.

Results: RQ4 – Value of Affordability

Full model

RMSE 9.13 points (.09131)

$R^2 = .76769$

Without affordability

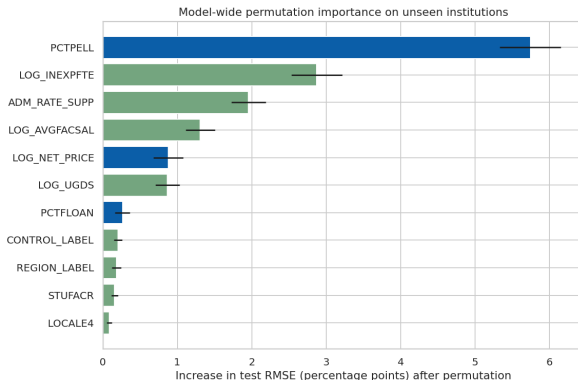
RMSE 11.19 points (.11192)

$R^2 = .65093$

$\Delta R^2 = .11675$

All five folds favored the full set.

Pell share had the largest permutation importance.



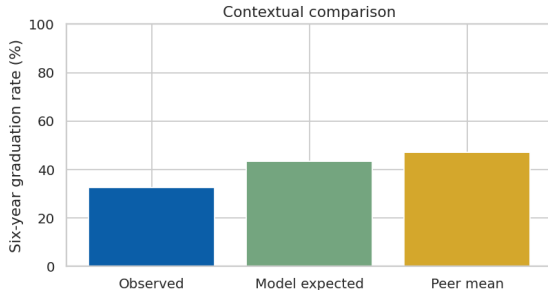
Results: Uncertainty and Reliability

- ▶ CV+ 95% intervals achieved 97.90% holdout coverage; average width was 44.04 points.
- ▶ Metrics are reported only for subgroups with at least 20 test institutions.
- ▶ Rural RMSE was 11.94 points versus 9.13 overall; for-profit subgroup metrics were not reported because the test subgroup contained 15 institutions.
- ▶ Complete-case, percentile-clipping, and alternative-outcome checks retained R^2 from .738 to .768.
- ▶ The point prediction can frame a review question, but it cannot classify one college with precision.

Uncertainty and sensitivity analyses from the executed notebook.

Implementation and User Benefit

- ▶ Select an institution and compare observed, model-expected, and peer rates.
- ▶ Keep the 95% interval, peer rule, peer count, source date, and model version visible.
- ▶ Investigate large gaps with approved local evidence on credits, courses, balances, advising, or stop-out.
- ▶ Do not use the output for rankings, admissions, funding, causal simulation, or student-level scoring.



Example: Abraham Baldwin Agricultural College
32.51% observed | 43.32% expected | 47.12% peer mean
95% prediction interval: 20.86–65.99% | peer $n = 59$
Peers: control + locale + enrollment quartile
Release: June 10, 2026 | model: QM640-v1.0 | association, not causation

Conclusion

- ▶ Affordability adds adjusted and predictive information; Pell share provides the strongest affordability signal and the largest individual permutation importance.
- ▶ Adjusted completion differs by control, while locale differences are unresolved.
- ▶ Random forest performs well on unseen colleges, but RMSE and broad intervals limit individual decisions.
- ▶ The benchmark is appropriate for documented internal review, not causal claims or ordered rankings.

Limitations and Future Work

LIMITATIONS

- ▶ Cross-sectional institutional data support association and prediction, not causal intervention.
- ▶ Important factors remain unmeasured, including preparation, transfers, work hours, course access, and advising.
- ▶ Rural error is higher, for-profit evidence is sparse, and one frozen split limits generalization.

FUTURE WORK

- ▶ Use rolling-origin validation on later Scorecard releases to test temporal stability.
- ▶ Add targeted predictors only when their rationale, coverage, and missingness are acceptable.
- ▶ Develop group-conditional or adaptive intervals and retain model, release, and subgroup evidence in the audit panel.

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